

Maintenance Wizard Report

Diagnosis

Imminent or advanced stage of CCM Segment Bearing failure.

Root Cause

Likely due to lubrication failure (insufficient, contaminated, or incorrect lubricant), excessive loading, misalignment, or fatigue. Contamination ingress cannot be ruled out.

Risk Level

CRITICAL - High potential for immediate production disruption and secondary damage.

Failure Probability

VERY HIGH - Symptoms indicate imminent catastrophic failure of the bearing.

Remaining Useful Life

EXTREMELY LOW - Measured in hours, not days or weeks; failure is likely within current operating shift or very soon.

Catastrophic Risk

HIGH - Potential for strand breakout, segment deformation, damage to adjacent segments, caster stoppage, hot metal spill, and safety hazard.

Immediate Actions

['Notify Production/Operations management immediately regarding critical status.', 'Implement continuous, intensified monitoring of vibration, temperature, and noise on the affected segment.', 'Verify lubrication supply to the affected segment bearing; check pressure, flow, and lubricant level/quality.', 'Prepare emergency segment change crew and equipment.', 'Evaluate reducing casting speed if technically feasible and safe, to mitigate progression of damage.', 'Initiate an emergency shutdown/caster drain if conditions rapidly deteriorate or reach critical predefined limits (e.g., excessive vibration, rapid temperature spike).']

Short Term Actions

['Schedule and execute an expedited segment replacement during the next available maintenance window or forced outage.', 'Order and expedite delivery of required spare parts if not in stock (bearing, seals, housing components, segment assembly).', 'Post-failure analysis of the removed bearing and segment to precisely determine the root cause.']

Long Term Actions

['Review and optimize lubrication procedures, lubricant type, and frequency for all CCM segment bearings.', 'Upgrade condition monitoring systems (e.g., online vibration analysis, acoustic emission sensors, thermal imaging) for proactive detection across all segments.', 'Investigate potential for ingress of scale, water, or other contaminants into bearing housings and implement protective measures.', 'Evaluate bearing design, material, and supplier performance for durability and suitability.', 'Implement comprehensive training for maintenance personnel on proper bearing installation, lubrication, and segment handling.']

Spare Procurement

['CCM Segment Bearing (specific type and size) - High priority.', 'Bearing seals, gaskets, and associated fastening hardware.', 'Lubricant supply (ensure correct grade and quantity).', 'Consider a complete spare CCM segment for quick replacement readiness and reduced downtime.']

Executive Summary

A CCM segment bearing is exhibiting critical symptoms of high vibration, increasing temperature, and abnormal noise, indicating imminent failure. This poses a severe risk of catastrophic production loss, secondary equipment damage, and safety hazards. Immediate intervention, including preparedness for an emergency segment change, is paramount.

Maintenance Recommendation

Perform an immediate segment change-out at the earliest safe opportunity. Conduct a thorough forensic analysis of the failed bearing to identify the precise root cause and implement corrective measures across the entire CCM fleet to prevent recurrence.

Conclusion

The current state of the CCM segment bearing demands urgent attention to prevent a catastrophic failure. Prompt replacement is critical, followed by a detailed root cause analysis and the implementation of long-term preventative measures to ensure reliable operations and minimize future risks.